

Usability Evaluation

Overview

This usability evaluation was conducted to measure and document the usability and overall user experience (UX) of the high-fidelity prototype for the "Laylah" Platform". The evaluation primarily focused on assessing the efficacy, efficiency, and success rate of users across various demographic age groups while interacting with the platform's interfaces and executing its core tasks.

Evaluation Objectives

- **Measure Efficiency & Effectiveness:** Verify the users' ability to independently complete the platform's primary user flows with minimal friction.
- **Identify Usability Friction Points:** Uncover and diagnose systemic usability flaws and interface design deficiencies that negatively impact user interaction.
- **Assess User Satisfaction:** Quantify subjective satisfaction and psychological comfort experienced by users during system interaction.

Methodology & Implementation

- **Participants & Target Audience:**
 - A representative sample of 8 participants was selected from the platform's target demographic (individuals engaged in wedding planning activities). The sample was stratified across distinct age cohorts to ensure comprehensive and inclusive insights:
 - Youth Cohort (Ages 21–29): 1 Male + 1 Female
 - Cohort (Ages 30–39): 1 Male + 1 Female
 - Cohort (Ages 40–49): 1 Male + 1 Female
 - Older Adult Cohort (Ages 50–59): 1 Male + 1 Female
- **Testing Environment & Apparatus:**
 - **Testing Method:** In-person Moderated Usability Testing.
 - **Rationale:** This approach allowed the moderator to closely observe non-verbal behavioral cues, detect subtle cognitive

hesitation, and provide immediate post-task probing, which is particularly critical for capturing the interaction patterns of older demographics.

- **Environment & Hardware:** Sessions were conducted in a controlled, quiet environment simulating realistic usage conditions. The prototype was deployed and evaluated on mobile devices.
- **Think-Aloud Protocol:** Participants were continuously encouraged to verbalize their thought processes, emotional states, and perceived friction points in real-time, enabling precise tracking of cognitive breakdowns.

Method setting and Execution plan

Execution Plan: Step-by-Step Procedure

- **Step 1 (Briefing & Orientation):** The participant receives a high-level briefing regarding the evaluation goals and structural guidelines.
- **Step 2 (Task Execution):** The participant is instructed to execute three predefined tasks completely independently, without external guidance.
- **Step 3 (Observation & Data Logging):** The researcher monitors user interactions non-intrusively, logging behavioral patterns, errors, and quantitative metrics.
- **Step 4 (Post-Task Survey):** The participant completes a brief quantitative questionnaire to assess subjective satisfaction and perceived ease of use.
- **Step 5 (Data Analysis):** Quantitative data is aggregated to compute success rates, average completion times, and error frequencies.

Predefined Evaluation Tasks (The 3 Tasks):

- Task 1: Create a new user account on the platform and successfully log in.
- Task 2: Navigate through the available services, select a specific vendor service, and complete the booking process.
- Task 3: Navigate to the package customization module, design a personalized package, confirm the order, and reach the final confirmation screen.

Evaluation methods

Execution Script Read to the User

"Welcome, and thank you for participating in our study today. I would like you to perform three specific tasks on this application. First, please create a new account and log in. Second, browse the available services and book a service of your choice. Third, navigate to the packages section, customize a personalized package, and confirm the order. As you navigate, please think out loud—express what you find intuitive and what you find confusing. I am here only to observe and take notes, and I will not be able to assist you. You may begin now".

Success Criteria & Metrics

To objectively evaluate user performance across the entire session, the following benchmark criteria were established:

- **Task Success Levels:**
 - **Full Success:** The participant completed all core tasks accurately and within the established time limit.
 - **Partial Success:** The participant successfully completed the workflow and tasks but exceeded the designated time threshold.
 - **Failure:** The participant abandoned the tasks, requested moderator assistance, or failed to complete the core user flow.
 - **Time-on-Task Threshold (Time Limit):** A maximum ceiling of 5 minutes to achieve Full Success.
 - **Error Tolerance Threshold (Error Limit):** A maximum allowance of 3 critical or non-critical errors per user across the session.
- **Measured Variables & Metrics:**
 - The following quantitative and qualitative metrics were recorded: Participant Age/Gender, Task Success Level, Time-on-Task, Error Frequency, Satisfaction Score (Likert Scale 1–5), and Qualitative Feedback/Behavioral Cues.

Data analysis and visualization

Data Recording Table

Participant ID	Demographic Cohort	Task Success Level	Time-on-Task (Minutes)	Error Frequency	Satisfaction Score (1-5)	Qualitative Observations & User Experience Notes
U1	Male (21-29)	Full Success	3.50	0	5	Executed tasks seamlessly with high interaction fluency; no observable friction.
U2	Female (21-29)	Full Success	4.50	2	3	Experienced cognitive friction on the vendor page due to an counter-intuitive information hierarchy (booking calls-to-action appeared prior to service details and imagery). Disorientation occurred within the package customization view; expected a page transition or auto-populated list beneath the budget field, failing to initially notice the "Add Service" call-to-action.
U3	Male (30-39)	Failure	6.00	2	2	Encountered data-entry validation failure due to Google Autofill inserting spaces into the phone field, prompting a disruptive manual country code

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						requirement. Optional input fields lacked distinct visual labeling, leading the user to treat them as mandatory. Failed to locate the obscured "Add Service" button, resulting in task abandonment and exceeding the session time limit.
U4	Female (40-49)	Full Success	3.40	0	4	Stable user experience; completed all workflows successfully within the acceptable timeframe.
U5	Male (40-49)	Partial Success	6.00	2	3	Attempted to manually input the date via text input instead of utilizing the date-picker icon; text fields cleared unexpectedly, inducing user frustration. Session duration exceeded the limit (6.00 mins) but tasks were eventually completed.
U6	Female (50-59)	Full Success	3.50	1	4	Suffered temporary disorientation within the package customization workflow; anticipated a multi-step wizard transition or service listing beneath the

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						budget input before identifying the functional layout of the "Add Service" button.
U7	Male (50-59)	Full Success	2.50	1	4	The "Add Service" call-to-action suffered from poor visual dominance and was not immediately perceived during initial scanning.
U8	Female (50-59)	Partial Success	7.00	1	4	Repeatedly tapped on the service display card container without any corresponding system feedback or navigation transition. Proved highly deliberate and slow, yielding an expanded completion time (7.00 mins) but successfully completed the tasks with low errors.

Quantitative & Behavioral Data Analysis

- **Full Success Rate**
 - **Success Rate:** 62.5%
 - **Average Time:** 25%
 - **Aggregate Success Rate (Full + Partial):** 87.5%
 - **Mean Time-on-Task (Average Time):** 4.55 Minutes
 - **Mean Error Frequency (Average Errors):** 1.13 Errors

Interpretation of User Performance & Behavior

- **Benchmark Evaluation:** Introducing the "Partial Success" metric reveals a more refined picture of system health. While an 87.5%

Aggregate Success Rate indicates high task completion viability, only 62.5% achieved Full Success within the 5-minute threshold. The fact that 25% of users (U5, U8) required extra time to bypass interface defects underscores that the current layout compromises user efficiency rather than task capability.

- **Correlational Patterns:** A direct positive correlation was observed between "element visual obscurity" and "inflated task completion times." Users who struggled to locate the "Add Service" button (e.g., **U2, U3, U6**) or encountered input validation rigidity (e.g., **U3, U5**) exhibited a substantial, linear increase in completion times, dropping them from Full to Partial Success.
- **Anomalous Behavioral Insight (User U7 vs. User U8):** A highly compelling behavioral anomaly occurred within the older adult cohort (Ages 50–59). User (U7) achieved the absolute fastest completion time across the entire sample at (2.50) minutes (Full Success) with only one error, directly refuting the assumption that older demographics inherently struggle with digital literacy. When contrasting (U7) with (U8) who also committed only a single error but marked a Partial Success at (7.00) minutes, a deeper demographic interaction paradigm emerges. Older adults heavily rely on a "Step-by-Step Deliberate Navigation Style". Unlike younger cohorts who engage in predictive fast clicking (which often triggers validation errors), older users carefully read on-screen text and execute actions linearly to prevent mistakes. This extreme caution is empirically proven by (U8)'s prolonged 7.00-minute session; she prioritized accuracy over speed. This same linear approach allowed (U7) to seamlessly bypass the hidden system pitfalls—such as the autofill formatting bugs that caused the total failure of younger user (U3) resulting in rapid execution when no sub-surface code blocks interfered.

Usability Findings & Diagnostic Breakdown

- **Finding 1:**
 - **Problem Statement:** Critical visual obscurity of the "Add Service" call-to-action within the package customization view, breaking the natural user flow.

- **Empirical Evidence:** Explicitly verbalized as a primary bottleneck by (U2, U3, U6, and U7). This deficiency directly caused the task failure of (U3) (6.00 minutes), who abandoned the flow due to a total loss of directional orientation .
- **Root-Cause Analysis:** The interface design isolates the primary action button below the fold and beneath the budget field rather than utilizing an automated, progressive disclosure model or inline scroll feed. This arrangement directly violates users established mental models regarding configuration wizards.
- **Finding 2:**
 - **Problem Statement:** Severe rigid constraints and error-handling vulnerability within data entry forms (Phone & Date fields) coupled with an absence of inline field affordances.
 - **Empirical Evidence:** User U3 experienced severe friction when the system rejected standard Google Autofill syntax due to white-space formatting and forced manual country code rewriting. User (U5) generated 2 consecutive errors and dropped to Partial Success (6.00 minutes) when his manual text entry vanished inside a rigid date field that lacked explicit visual text wrappers or "(Optional)" tags.
 - **Root-Cause Analysis:** Absence of front-end data sanitization scripts capable of stripping white spaces from strings automatically, combined with utilizing date inputs that lack dual keyboard/calendar entry flexibility.
- **Finding 3:**
 - **Problem Statement:** Poor layout-driven information hierarchy on vendor profiles and lack of touch-target click ability on service card components.
 - **Empirical Evidence:** User U2 experienced layout disorientation due to actions appearing before informational context. User U8 spent a major portion of her 7.00-minute session repeatedly tapping service card illustrations without any interactive feedback from the system.
 - **Root-Cause Analysis:** The UI elements violate standard visual hierarchy principles by positioning secondary scheduling forms above core service descriptions and media

galleries. Furthermore, the card container was designed as a static graphical background rather than a fully interactive card wrapper link.

Proposed Solutions

#	Proposed Solution	Cause and Development	Impact	Priority
1	Redesign the "Add Service" element as a visually dominant primary button, or transition the layout to an auto-populated, inline stream beneath the budget field.	Eliminates cognitive disorientation and ensures fluid continuity across the package customization user journey.	Completion Time / Success Rate	Critical (High)
2	Wrap the entire service card container into a unified, clickable interactive target, incorporating distinct, well-spaced buttons differentiating "View Details" from "Direct Booking."	Optimizes layout navigation, establishes clear clickability affordances, and mitigates errors stemming from tight interactive element clustering.	Completion Time / Error Frequency	High
3	Implement front-end string sanitization to auto-format phone numbers, deploy an input mask supporting simultaneous manual typing and date-picking, and explicitly apply text labels with "(Optional)".	Lowers form-validation friction during registration and checkout, significantly accelerating conversion speeds.	Error Frequency / Completion Time	Medium

Strategic Design Decision

- **Primary High-Impact System Overhaul:** The development team will immediately restructure the package customization interface architecture. The workflow will be modified to display available service line-items dynamically and automatically based on user-defined budgets, while converting the current ambiguous "Add Service" button into a persistent, contextual action item.
- **Alternative Design Concept Rejected:** Complete elimination of the custom package-building module in favor of offering rigid, pre-configured vendor bundles exclusively.
- **Rationale for Rejection:** Removing this feature would directly compromise the platform's core unique selling proposition (USP),

which centers on providing users with total planning flexibility and customization freedom. Forcing users into fixed bundles would severely degrade long-term customer adoption.

- **Data-Driven Justification:** User intent for customization was heavily reinforced by qualitative data. Despite severe interface friction, 6 out of 8 users actively and vocally persisted in trying to complete the custom package flow. Furthermore, user satisfaction plummeted sharply to scores of (2) and (3) only when encountering layout barriers preventing flexible customization (e.g., U2, U3).

Evaluation Limitations & Analytical Impact

- **Structural Limitation:** The study is bounded by a relatively small usability sample size (8 participants) and was conducted as an in-person moderated laboratory evaluation rather than a distributed, unmoderated remote study in the users' native, everyday environments.
- **Analytical Impact of Limitation:** These constraints restrict the broad generalizability of absolute completion times. The physical presence of a moderator in an active logging session naturally introduces a minor Hawthorne Effect—where users experience elevated performance anxiety. This anxiety may artificially inflate time-on-task metrics or cause unnatural hesitation in form inputs compared to organic, unobserved platform usage conditions.

Initial User Research: Data gathering Processes

Initially, projections and preliminary analysis were used to identify the target user group by predicting who might benefit from the platform and their potential needs. This provided an initial framework to guide the research process.

Subsequently, a set of survey questions was designed based on these assumptions to gather preliminary data that would help in better understanding the users. These questions were then piloted with a number of individuals, and their feedback was used to refine and improve the wording of the questions to ensure their clarity and effectiveness.

The analysis of the survey results revealed four main personas: the persona with a specific goal, the persona without experience, the persona for comparison, and the persona who is busy.

The survey and interview result also showed that approximately half of the participants found the topic of marriage and its preparations stressful and exhausting due to the complexity of the procedures and the abundance of details. This highlights the importance of the platform in simplifying and facilitating the process. Participants expressed their desire for any solution that saves effort and time and makes the experience easier and more comfortable. Several indicated that relying on online platforms has become a preferred option because it is more convenient and easier compared to traditional methods.

The results also showed that users' behavior today is increasingly reliant on the internet and technology. The platform helps those lacking experience in wedding preparations, alleviates the burden on those overwhelmed by wedding arrangements, and generally simplifies the wedding preparation process, aligning with the broader trend toward digital transformation. Similar to delivery and hotel booking apps, reliance on electronic solutions has become commonplace, with most services now being completed easily and quickly online.

Data gathering Techniques Description

Data collection was conducted through direct interviews with participants, both face-to-face and remote online. The survey questions were integrated into these interviews as part of the interactive dialogue, rather than being presented as a traditional form.

This approach was adopted to ensure accurate and informed responses, as direct interaction allows participants greater freedom of expression, especially in open-ended questions. This approach also helps reduce the likelihood of random or inaccurate responses, which can occur in traditional surveys due to their sheer volume or a lack of participant engagement.

In addition, the interviews contributed to extracting unexpected insights and perspectives, enhancing the quality of the qualitative data alongside the quantitative data. Care was also taken to ensure the diversity of the participating sample in terms of gender (male and female) and age groups, and by combining in-person and remote interviews, to achieve greater comprehensiveness in the results.

Consequently, the interview-questionnaire approach contributed to collecting rich and high-quality data, resulting from direct interaction with participants and a focus on their experiences, rather than simply completing a traditional questionnaire.

Method setting and Execution plan

The execution of the data collection process was carried out by the project team, where each responsible member conducted interviews and questionnaire-based data collection directly with at least three participants. The team members were responsible for asking the questions and accurately recording responses, either through online calls or in-person field interviews, rather than simply distributing a survey link.

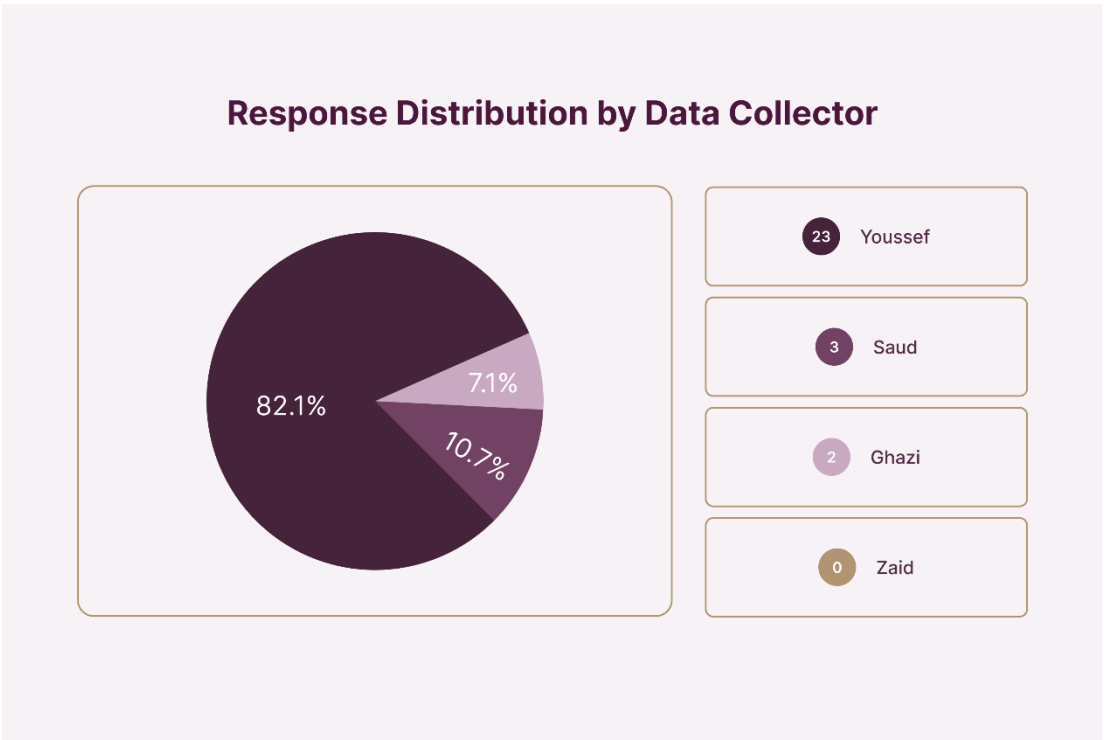
The sample selection was intentionally structured to ensure diversity across different age groups as much as possible. Participants were targeted in the following categories:

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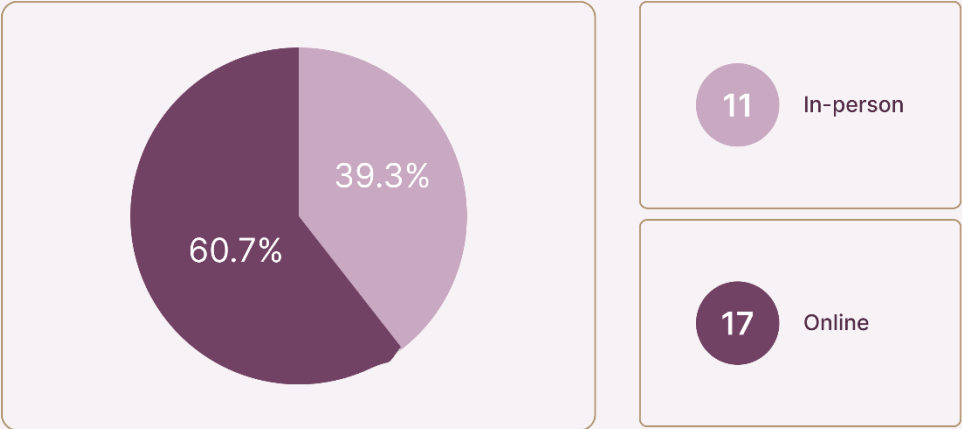
- 29–21years old
- 39–30years old
- 49–40years old
- 50-59 years old

The data collection process was conducted over multiple days and in different locations within Makkah, combining both field interviews and remote sessions. Participants were selected from diverse backgrounds, including different cities and cultural backgrounds within Saudi Arabia, to ensure broader representation and more reliable results.

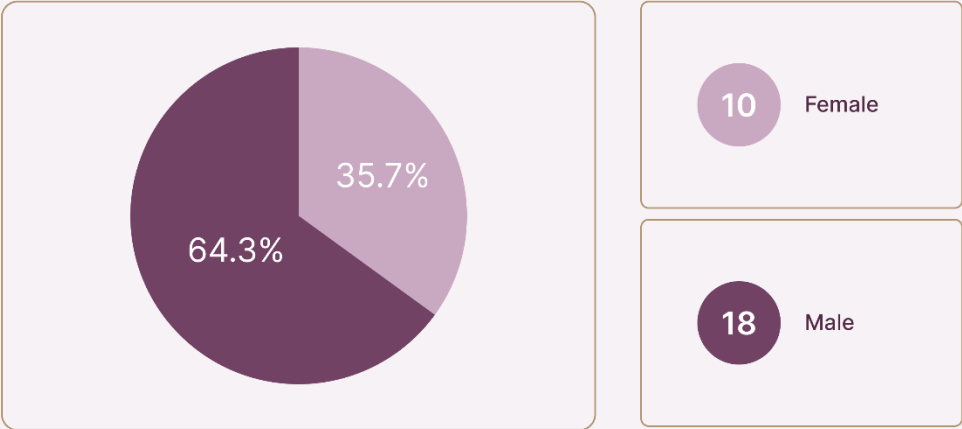
Data analysis and visualization



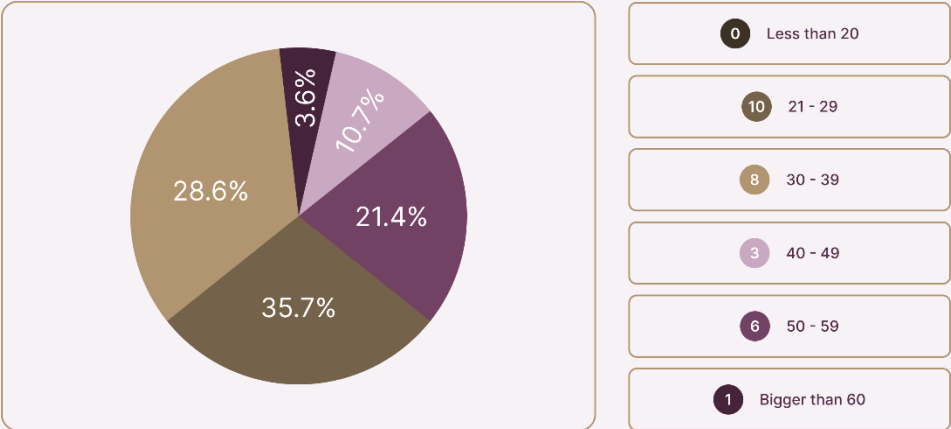
Data Collection Channels



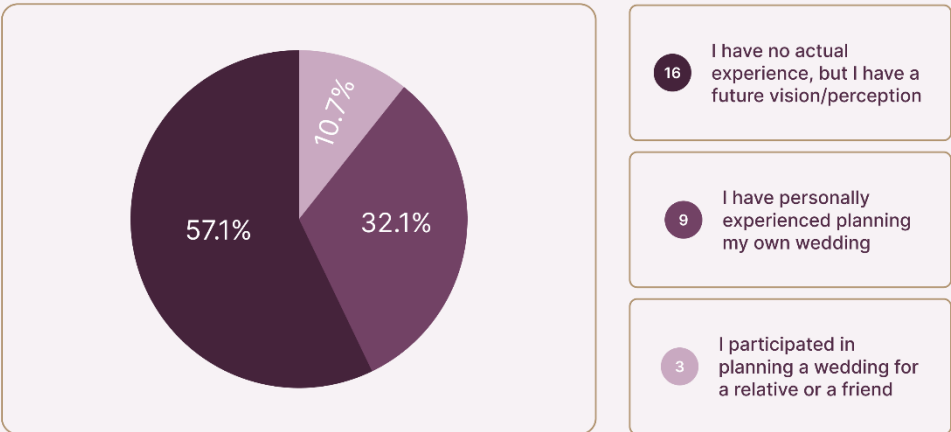
Users Gender



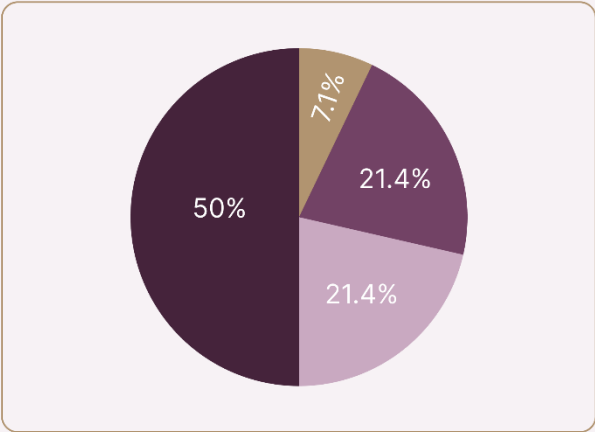
Age distribution of users



The nature of the participants' experience with weddings

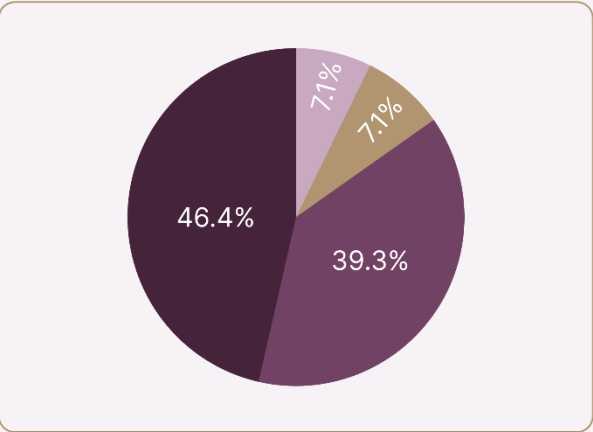


Expertise Level Distribution



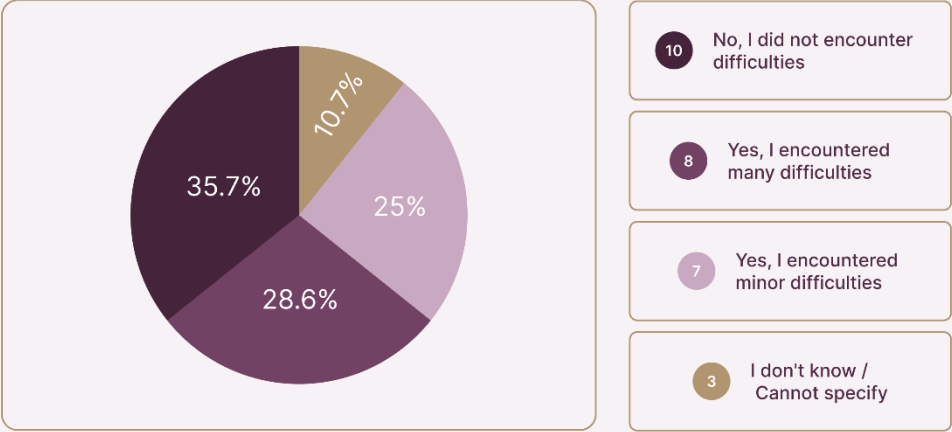
- 14 Expert: I have extensive experience and can efficiently organize the event
- 6 Intermediate: I have good experience and can handle most matters
- 6 Beginner: My experience is limited and I need assistance or supervision
- 2 Not applicable to me

Rating of Wedding Services Search Experience

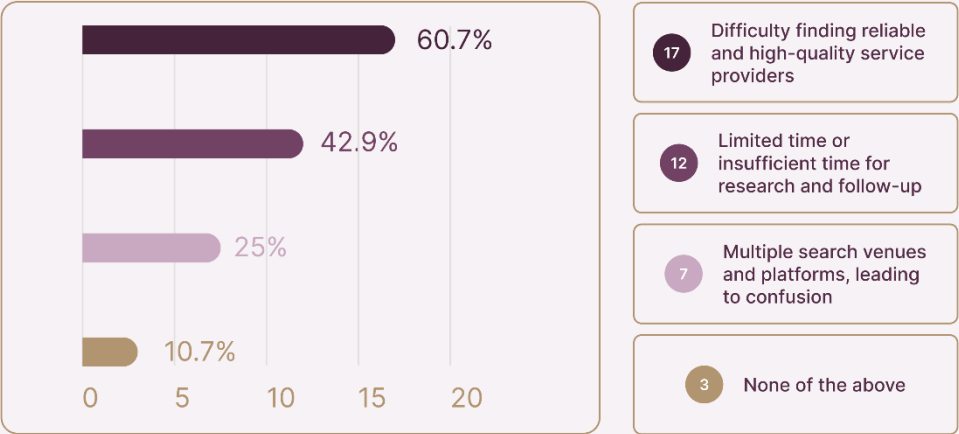


- 13 Complex and stressful
- 11 Tiresome and time-consuming
- 2 Acceptable
- 2 Easy and enjoyable

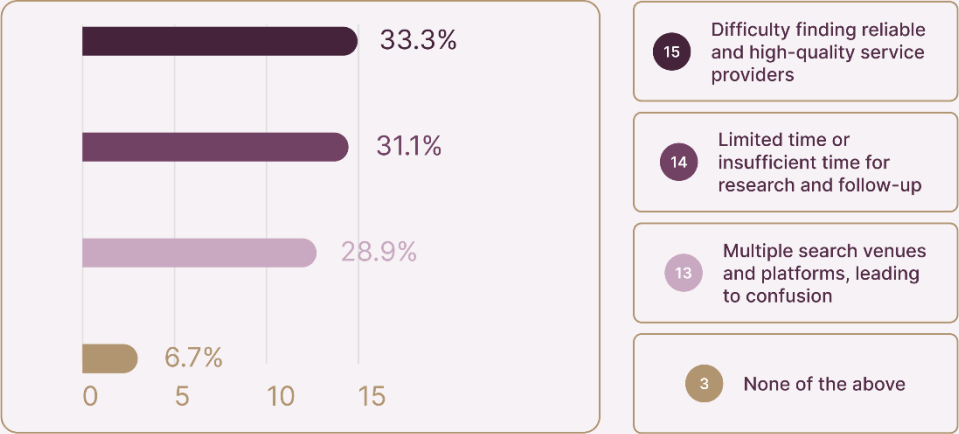
Extent of Problems Faced in Searching for Services



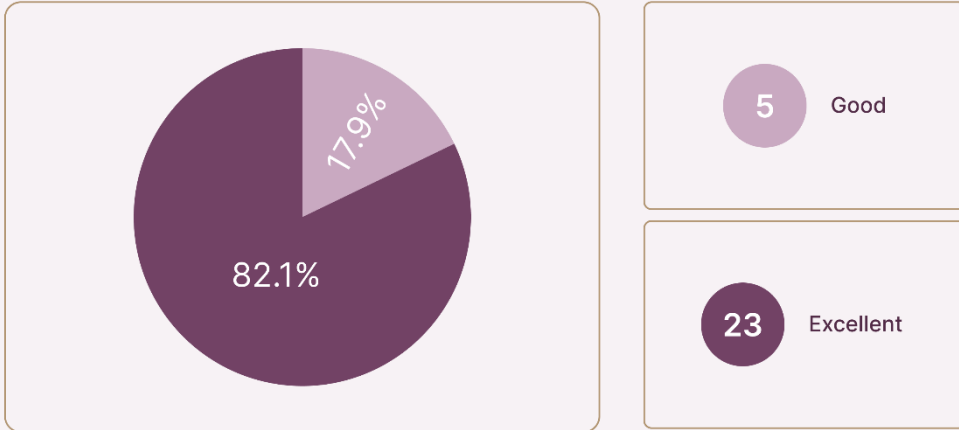
Sources of Stress in Wedding Plannin



Decision-Making Methods for Wedding Services



Acceptability Level of the Platform Idea



Expected Usage Behavior for the Platform

